

## Engineering Design Report | Car #90: Brunel Masters Motorsports 2026 FS CC

*Brunel University London — College of Engineering, Design and Physical Sciences*

Brunel Racing's 2026 Concept Class entry is a single-motor, rear-wheel-drive electric vehicle designed as the engineering foundation for the running FS Class car for the 2027 season. Vehicle mass target: 260–280 kg without driver. Peak motor power 80 kW per EV2.2.1 (motor rated 124 kW, derated to 73 kW peak). Accumulator capacity 8.16 kWh at 388.8 V nominal.

### A) Overall Vehicle Concept

The vehicle is structured around three weighted design drivers, namely manufacturability, reliability, and cost, with peak performance treated as the consequence of well-executed engineering within these constraints rather than as the starting point. This ordering reflects the team's available resources, workshop capability, supplier relationships, prior experience, and the explicit constraint that the concept must progress to a built and tested vehicle within twelve months.

The choice between IC, EV, and hybrid powertrains was the first concept-level decision, and a weighted decision matrix was applied to resolve it. EV was ultimately selected on the basis of several drivers, including the global automotive industry's transition to electric powertrains, which makes Concept Class progression to a professional-grade EV team in 2027 a defensible direction; the supportive FS UK regulatory environment for EVs, which now includes the removal of the noise test, an established base of motor and inverter suppliers, and charging infrastructure provided at competitions; and the suitability of a single-motor RWD layout in avoiding the dual-motor complexity that would otherwise compound the EV learning curve. The trade-off accepted in making this choice is that the HV safety burden, encompassing training, shutdown circuit implementation, isolation monitoring, and cabling management, is substantially larger than its IC equivalent but is considered necessary.

The vehicle targets a mass of 260 to 280 kg without driver, and this upper-range Concept Class target is intentionally conservative so as to provide margin against detail-design growth. Within the EV architecture, a single-motor rear-wheel-drive layout was chosen over dual-motor and four-wheel-drive alternatives on reliability grounds, since it yields a simpler shutdown circuit, fewer HV components, fewer fault paths, a lower Design Verification Plan burden, and easier handover to the 2027 FS Class team. The accepted trade-off is the absence of torque-vectoring and the limitation of rear-axle traction by the open differential, both of which are considered acceptable at Concept Class scope.

Aerodynamic surfaces are limited to an undertray-diffuser and sidepods, with no front or rear wings fitted. At FS speeds, where the peak vehicle speed is approximately 95 km/h at 5,500 RPM motor speed with a 4.0:1 final drive and 411 mm tyre OD, the cornering-downforce gain offered by wings is outweighed by the mass, drag, CG-height, yaw-inertia, and lead-time penalties they would impose. The undertray-diffuser provides 130 to 180 N of downforce at 80 km/h with 50 to 55% front-biased loading, which maintains stable rear balance at the cost of lower peak grip than a wing-equipped car would achieve.

The target customer is the amateur club-level Formula Student-derived motorsport entrant, who values predictable handling, parts availability, and serviceability over peak performance. Major components have therefore been selected from established motorsport suppliers, including AP Racing brakes, Hoosier tyres, the EMRAX motor, the Cascadia inverter, and Molicel cells. Every component is OEM-purchasable, which reduces supply-chain risk and supports field maintenance during race weekends.

SMART performance targets include 0 to 100 km/h acceleration in under 4.5 s, with 3.97 s achieved in SiL; 1.5 g lateral cornering drawn from tyre  $\mu$  and the undertray contribution; an endurance energy budget of no more than 8.16 kWh across the 22 km circuit; and a Concept Class points target of 300 out of 460.

### B — Structural Design, Suspension and Brakes

**Frame:** The primary structure is a tubular spaceframe in AISI 4340 high-strength steel, TIG-welded with ER80S-D2 filler, preheated to 150 °C and post-weld stress relieved. The target frame mass is 47 kg including brackets and paint, while FEA returned a torsional stiffness of 2,922 Nm/deg against a

minimum target of 2,000 Nm/deg. The selection of 4340 over 4130 was driven by supplier stock availability and lead-time considerations, since 4340 was available pre-cut with a two-week delivery while 4130 was quoted at six weeks. Preheating and post-weld heat treatment are both within current workshop capability. The accepted trade-off is on mass, given that steel is denser than both aluminium and CFRP, and this is exchanged for manufacturability and reliable correlation between FEA predictions and the physical structure.

*Impact Attenuator:* The student-designed trapezoidal frustum is formed from AA6061-T6 with a 1.5 mm wall thickness and dimensions of 268.6 by 230 by 270 mm. Energy absorption from non-linear FEA is 8.45 kJ, which sits 15% above the 7.35 kJ Standard IA Reference threshold, and quasi-static crush testing of a representative sample is planned for the FS Class build phase.

*Suspension:* The suspension is a double-wishbone arrangement with pushrod actuation at both front and rear. Pushrod actuation was chosen for packaging reasons, since the front layout clears the steering rack while the rear permits lower accumulator mounting and consequent CG reduction. The motion ratio is 0.75:1 at both axles, giving wheel rates of 22.2 N/mm at the front and 36 N/mm at the rear, and roll rates of 279 Nm/deg at the front and 452 Nm/deg at the rear. Sprung-mass natural frequency targets are 3.18 Hz at the front and 3.3 Hz at the rear, with static camber set at  $-0.5^\circ$  at the front and  $-1.2^\circ$  at the rear, and static roll-centre heights of 24.6 mm at the front and 30.7 mm at the rear. Anti-dive is 10% and anti-squat is 15%, and adjustability is provided through shim stacks for camber, tie-rod length for toe, and spring perch position for ride height. The trade-off of the pushrod-and-bellcrank arrangement is that it adds corner mass relative to a direct-acting damper layout.

*Tyres and Brakes:* Hoosier 16.0 by 7.5-10 R25B tyres are mounted on 6-inch aluminium rims, which prioritises unsprung mass reduction at the cost of requiring warm-up management for the R25B compound. The braking system is a dual-circuit hydraulic four-wheel arrangement compliant with T6, using AP Racing CP2623 master cylinders, CP4226 four-piston calipers, and 170 mm by 4 mm AISI 420 ventilated stainless steel rotors. At 1g deceleration, brake pressures are 60 bar at the front and 40 bar at the rear, with a pedal force of 0.5 kN. A brake-over-travel switch interlocks the shutdown circuit, and the brake light operates per T6.3.1. Regenerative braking is integrated by pedal-travel partitioning in accordance with the rules, such that regen is permitted during the first 90% of pedal travel while the remaining travel directly actuates the hydraulics. The trade-offs accepted are that the dual-circuit hydraulic arrangement constrains regen sophistication, and that the 170 mm rotors are limited by the 10-inch wheel package, which reduces peak thermal capacity.

*Cockpit and Driver Controls:* A two-pedal box with  $\pm 50$  mm of adjustment is fitted. The APPS uses dual independent rotary position sensors with dual return springs in compliance with T11.8 and EV2.3, and the VCU plausibility logic cuts motor torque to zero on sensor disagreement greater than 10% sustained for more than 500 ms. The steering wheel is a 260 mm aluminium tube frame with quick-disconnect to T2.4, and the seat is a 6061-T6 aluminium monocoque with polyurethane foam padding.

## **C — Powertrain**

The powertrain is a single-motor rear-wheel-drive arrangement comprising an EMRAX 228 MV axial-flux permanent-magnet synchronous motor, a Cascadia Motion PM150DZ inverter, and a 540-cell Molicel INR-21700-P42A lithium-ion (NMC chemistry) accumulator configured as 108S5P at 388.8 V nominal and 8.16 kWh.

*Motor and Drivetrain:* The EMRAX 228 MV was selected against the EMRAX 188, 268, and the AMK DD5. Its peak torque is 220 Nm to S2 2-minute duty per the datasheet, with 112 Nm continuous, while its rated 124 kW peak at 5,500 RPM is operated at 80 kW in accordance with EV2.2.1, and the continuous figure is derated to 73 kW to maintain thermal headroom. The base speed at the maximum torque point is 4,685 RPM. The 188 was insufficient at the 260 to 280 kg vehicle mass, the 268 added 6 kg of mass for capability that would have gone unused under the power cap, and the DD5 would have required additional transmission. Speed sensing is provided by a TAMAGAWA TS2620N21E11 resolver.



A single-speed 4.0:1 reduction was chosen over a two-speed arrangement on simplicity grounds, eliminating shift logic and solenoid failures. MATLAB simulations were used to optimise the ratio so that the vehicle reaches 95 km/h at the peak power point of 5,500 RPM, which is appropriate for both FS sprint and endurance phases. The drivetrain includes an open differential together with half-shafts of 4140 steel at 25 mm OD with splined tripod CV joints. The trade-off of the open differential is the rear-axle traction limit at corner exit, which is accepted both for the saving of approximately £3,000 and for the simplified compliance route.

*Inverter, Accumulator and Control:* The Cascadia PM150DZ inverter was selected for its 150 kW peak and 87 kW continuous discharge envelope, which matches the motor and accumulator; its 800 V bus headroom over the 453.6 V maximum accumulator voltage; its integrated CAN 2.0B interface; and its IP66 motorsport-rated enclosure. Vector-controlled torque demand is implemented with software de-rating to 80 kW, and the regen function is firmware-rate-limited so as to preserve rear-axle brake balance.

The accumulator's 108S5P configuration provides a nominal voltage of 388.8 V and a maximum of 453.6 V, with pack energy of 8.16 kWh derived from 540 cells at 3.6 V and 4.2 Ah. Continuous discharge is 225 A, drawn from 5P at 45 A per cell as specified in the Molicel datasheet, giving 87.48 kW continuous and 102 kW peak. Charging is at 9.5 kW power, completing a full charge in 1.5 hours at 95% efficiency. Per EV5.3.2, the pack is divided into six 18S5P segments at 64.8 V nominal and 75.6 V maximum each, with each segment storing 1.14 MJ of energy, which sits under the 6 MJ ceiling, while the 75.6 V remains under the 120 V limit and the 6.3 kg cell mass remains under the 12 kg ceiling when the container is included. All segment criteria are therefore satisfied with design margin in hand.

The Vehicle Control Unit is a dSPACE MicroAutoBox II running Simulink firmware at a 100 Hz update rate. A Stateflow state machine handles the start-up sequence of pre-charge, then AIRs closing, followed by shutdown circuit verification of the IMD, AMS, BSPD and BOTS, and finally entry into Ready-to-Drive. SiL validation has passed 18 of 21 Design Verification Plan tests, with the three remaining tests requiring Hardware-in-the-Loop. Measured response times are 10 ms or less for the SDC, 87 ms or less for IMD fault detection (which sits within the regulatory window), 68 ms or less for APPS failure handling, and 412 ms or less for brake override (which sits within the EV2.3 500 ms requirement).

Thermal management is handled by a single liquid loop comprising a Bosch pump operating at 8 to 10 LPM, a sidepod-integrated bar-plate radiator with 0.5 m<sup>2</sup> frontal area, a 2 L reservoir, and a 55 °C thermostat. Transient thermal FEA at 80 kW predicts a motor stator temperature of approximately 85 °C and an inverter junction temperature of approximately 90 °C, both of which sit within de-rating margins.

In terms of performance, the SiL 0 to 100 km/h time is 3.97 s against the 4.5 s target, endurance regen recovery is 9.4% (equivalent to 0.54 kWh across the 22 km circuit), and top speed is 95 km/h. The accepted trade-offs are that the HV burden is larger than the IC equivalent (which is considered necessary), that the accumulator adds 35 to 50 kg over an IC engine (which is accounted for in the mass budget), that the single-motor layout precludes torque-vectoring, and that the 80 kW cap limits the motor to 65% of its rated peak in exchange for a reliability margin over headline performance.

## **D — Timing, Manufacturing, Procurement and Team Organisation**

*Team and Procurement:* The team is organised into four engineering groups: Powertrain (led by Abdalla, with Viki as reduction-gear sub-lead), Chassis and Aero (Chara), Suspension (Kiarash), and Driver Controls (Jeffery). Coordination is provided through weekly leads meetings, a shared specifications spreadsheet, and the Technical Director (Sharan), who coordinates interfaces and CAD assembly.

The timeline runs from concept design between October 2025 and April 2026, into engineering and pre-submission between May and July 2026, then into procurement and fabrication between August and November 2026, followed by build and integration between December 2026 and March 2027,

and finally shakedown testing and tuning across March and April 2027. Long-lead items, including the EMRAX motor, Cascadia inverter, Molicel cells, custom uprights, and motor mount, have been flagged for order placement in August 2026 in order to clear the three-to-four-month lead-times.

*Risk Register, Top Three:* The first risk is that high-voltage training, qualification, and accumulator-handling supervision add both cost and lead-time, and this is mitigated through early HV-training booking together with a partnership with the university's electrical engineering safety officer. The second risk is that AISI 4340 frame welding requires preheating and stress relief, and this is mitigated by selecting a fabrication partner with documented 4340 capability rather than by attempting in-house welding. The third risk concerns the Concept-stage simulation gaps, namely the absence of a physical torsion test, IA crush test, thermal validation, and on-vehicle HiL, and these gaps are explicitly listed in the Design Verification Plan with progression milestones to closure ahead of FS Class submission.

*Knowledge Transfer:* Handover to the 2026 to 2027 cohort will be conducted through complete component-level documentation, archived CAD with full revision history, an engineering decision log, and a formal handover workshop scheduled for late August 2026.

## **E — Design for Manufacture, Profit, Customer Respect, Innovation, Lessons Learnt**

*Manufacturing Approach and Cost:* The vehicle is designed to be built within Brunel's engineering workshop with external support for chassis tube pre-cutting. Custom-machined components, including uprights, hub carriers, the motor mount, and pedal brackets, will be CNC-milled from 7075-T6 aluminium using university workshop time. Composite components, namely the BFRP and PVC-foam sidepods together with the CFRP and Nomex undertray, will be wet-laid-up by hand with vacuum bagging at room-temperature cure.

Cost-driven design choices include the open differential, which saves £3,000 at the cost of rear-axle traction; the BFRP sidepods, which save £1,800 against CFRP at the cost of a marginal mass increase; the 4340 frame, which costs the same as 4130 but was selected to remove lead-time risk through local stock availability; and the Hoosier R25B, which is the FS-standard tyre supported by a UK distributor and therefore removes supplier-risk exposure.

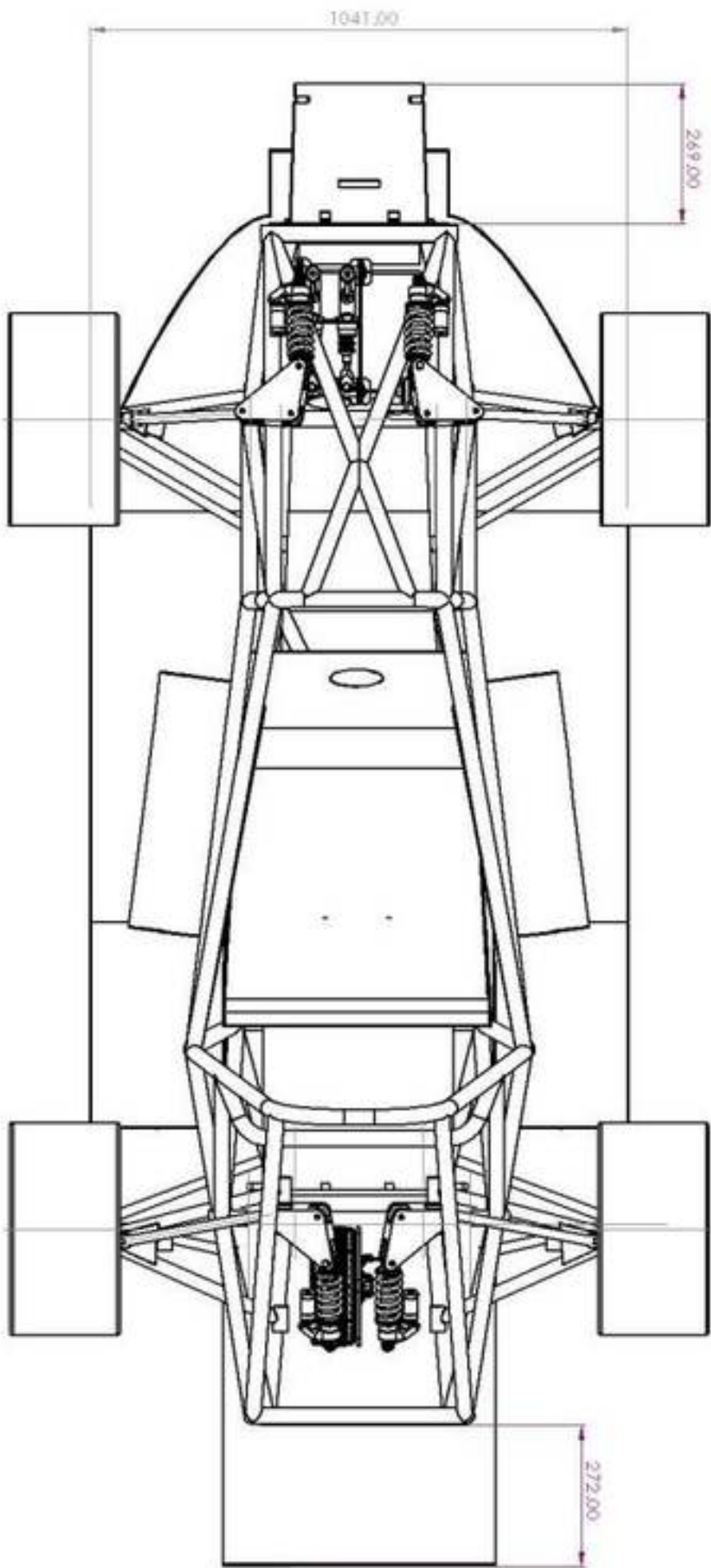
*Customer Respect and Innovation:* The target customer is the amateur club-level motorsport entrant, and this customer is served by component choices prioritising parts availability and field-serviceability, including AP Racing brakes, Hoosier tyres, the EMRAX motor, the Cascadia inverter, Molicel cells, and the AiM EVO5 data logger. Setup adjustability across pedal reach, steering reach, suspension camber, and brake bias is provided through conventional motorsport mechanisms requiring only standard tools, and there is no proprietary firmware or custom electronics that cannot be serviced.

Innovation has been deliberately limited, since for a first electric Concept Class entry established components and proven layouts represent a lower-risk path to credible FS Class progression than novel architectures would. The most notable design choice is the front and rear pushrod orientation differential, in which the front is oriented upward to clear the steering rack while the rear is oriented downward to lower the accumulator, yielding a modest CG benefit through layout discipline rather than through novel hardware.

*Lessons Learnt:* The first lesson is that cross-group decision discipline must be centralised early, since without a Technical Director coordinated specifications process the four sub-teams will converge on incompatible interface assumptions. The second is that HV scope is larger than initially scoped, since the shutdown circuit, AMS, IMD, BSPD, BOTS, pre-charge, latched reset, and TSAL collectively form a major workstream. The third is that mass-budget discipline must begin before component selection, since overruns of 15 to 20% accumulate easily without an itemised weekly-updated mass roll-up. The fourth is that long-lead components must be identified and order-prepared at the concept stage, since otherwise the detail-design-to-build timeline cannot close.

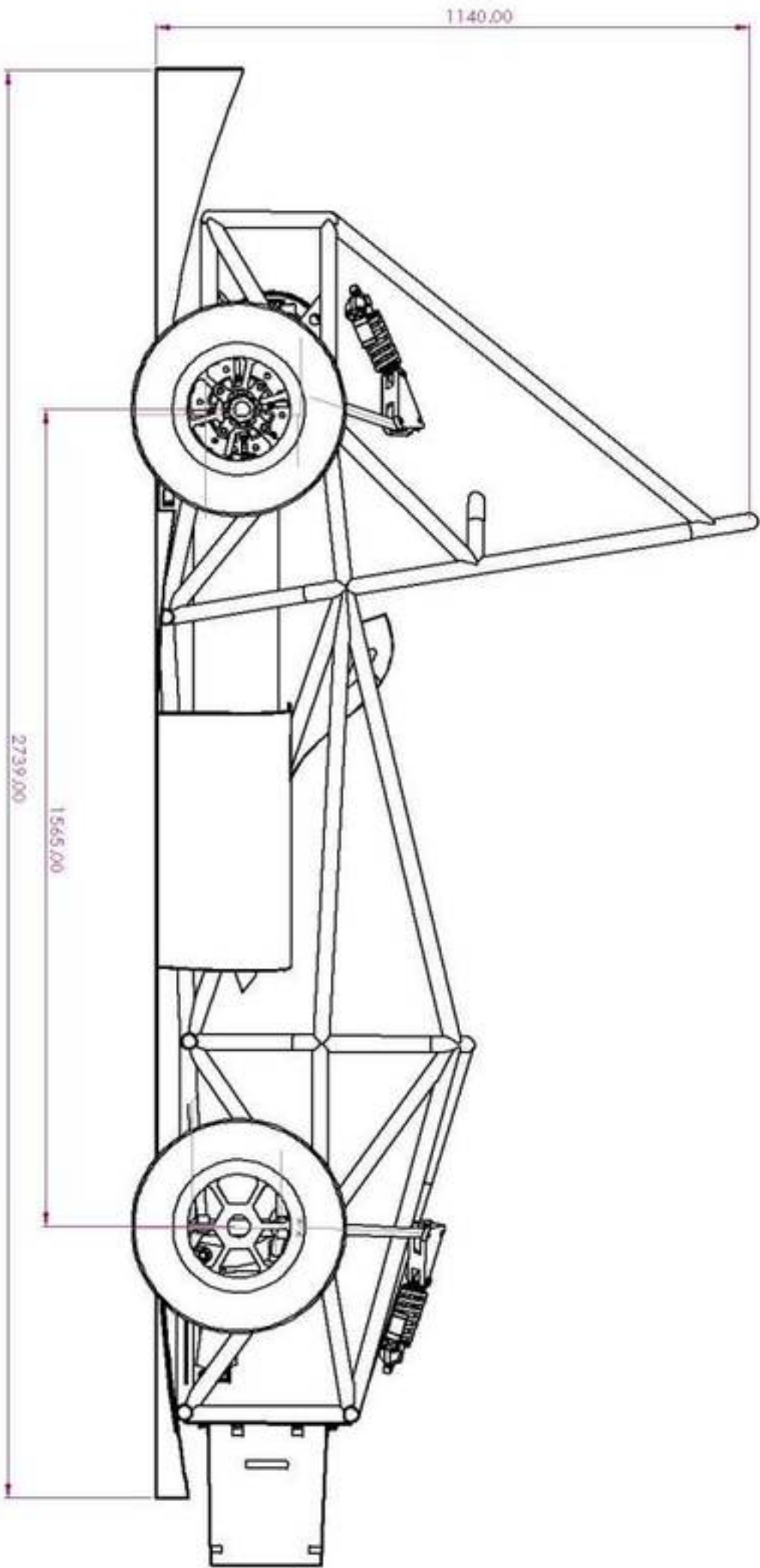


PLAN VIEW (TOP)



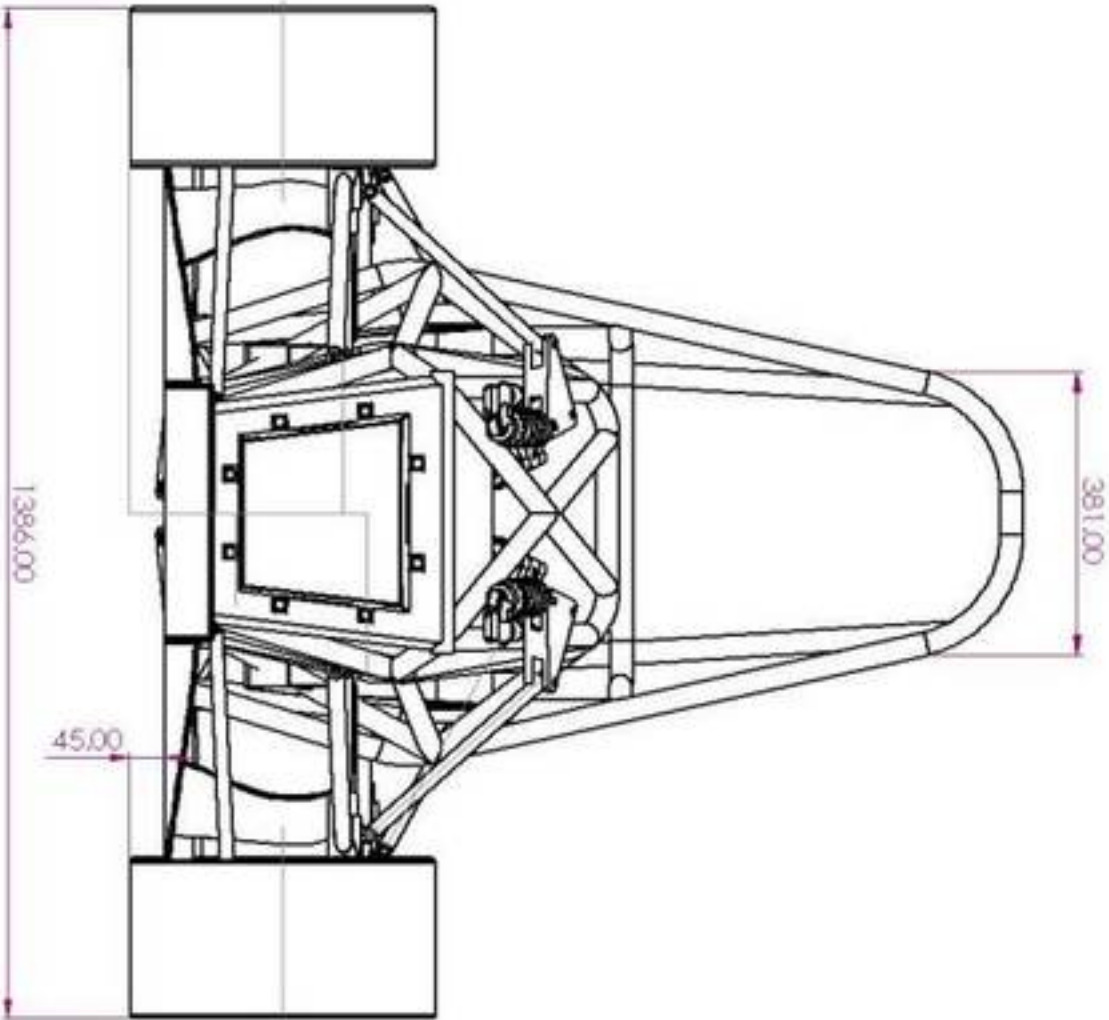
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| BRUNEL MASTERS MOTORSPORTS |  |
| RACE CAR #10 TOP VIEW      |  |
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SIDE VIEW



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| BRUNEL MASTERS MOTORSPORTS |  |
| TITLE: CAR PRO SIDE VIEW   |  |
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FRONT (END) VIEW



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| TITLE: CAR #10 SIDE VIEW   |  |
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